

Playwright with TypeScript

Week 5 – Classroom Activity



Program	Playwright with TypeScript
Module	TypeScript OOP Concepts
Week	Week 5
Activity	Implement Inheritance in TypeScript (Automation Testing Context)

1. Objective

To practice and understand inheritance in TypeScript by creating a parent class and child classes using a real-time automation testing scenario.

2. Problem Statement

Create a simple TypeScript program to demonstrate inheritance using a base class for common browser actions and child classes for specific actions used in automation testing.

3. Requirements

- Create a parent class called **BaseActions** with the following:
 - Property: `pageName` (string)
 - Methods: `openUrl(url: string)`, `click(locator: string)`, `getText(locator: string)`
- Create a child class called **LoginActions** that extends **BaseActions** with the following:
 - Additional properties: `usernameLocator` (string), `passwordLocator` (string), `loginButtonLocator` (string)
 - Method: `login(username: string, password: string)` that uses the parent class methods.
- Create another child class called **DashboardActions** that extends **BaseActions** with the following:
 - Additional property: `welcomeTextLocator` (string)
 - Method: `verifyDashboard()` that uses `getText()` from the parent class to validate the welcome text.
- Create objects for both **LoginActions** and **DashboardActions** and call their methods to demonstrate inheritance.
- Add meaningful comments and use proper TypeScript types.

4. Sample Test Data and Expected Output

Sample Test Data	Expected Output (Console)
<pre>// Create LoginActions object const login = new LoginActions("Login Page", "#username", "#password", "#loginBtn"); // Create DashboardActions object const dashboard = new DashboardActions("Dashboard Page", "#welcomeText"); // Test data const url = "https://opensource-demo.orangehrmlive.com/"; const username = "Admin"; const password = "admin123";</pre>	<pre>Navigating to URL: https://opensource-demo.orangehrmlive.com/ Clicked on: #loginBtn Login successful for user: Admin Navigating to URL: https://opensource-demo.orangehrmlive.com/ Dashboard welcome text: Welcome Admin</pre>

5. Technical Guidelines

- Use TypeScript (.ts file).
- Implement inheritance using the `extends` keyword.
- Define appropriate properties and methods in the parent and child classes.
- Use meaningful method names related to automation testing.
- Simulate actions using `console.log()` (no need to use actual Playwright code).
- Add comments to explain the code.
- Follow proper naming conventions and code formatting.

6. Learning Outcome

After completing this activity, you should be able to:

- Create a parent class and child classes in TypeScript.
- Use the `extends` keyword to achieve inheritance.
- Reuse parent class methods in child classes.
- Apply inheritance concepts in a real-time automation testing scenario.